

Development and Validation of a Novel Multiplex Cytokine Immunoassay on Magnetic Beads Using the Bio-Plex® Suspension Array System

BIO-RAD

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Abstract

Cytokines are key immunoregulatory proteins produced by white blood cells and other cell types. These low molecular weight proteins are involved in a wide variety of disease states, including inflammation and oncogenesis. We have chosen ten of the most widely publicized cytokines associated with the Th1/Th2 subsets (IL-2, IL-4, IL-5, IL-10, IL-12 (p70), IL-13, IFN- γ) and inflammation (IL-1 β , IL-6, TNF- α) to develop a multiplex immunoassay on the Bio-Plex suspension array system (based on Luminex xMAP technology). The validation of this sandwich immunoassay is described. This assay is unique in that it incorporates the use of magnetic beads and uses newly designed human serum and plasma standard diluents. The 10-plex cytokine assay was run in five independent assays for each standard diluent, and validated for standard curve accuracy (80–120%), precision (intra $\leq 10\%$ CV; inter $\leq 15\%$ CV), sensitivity (≤ 1 pg/ml), specificity ($\leq 1\%$ cross-reactivity), and parallelism ($< 10\%$ slope difference) using three independent donors for human serum and plasma samples, respectively. Assay robustness was determined by 96-well plate uniformity (3–7% CV) tests. For population analysis, cytokine levels were determined in 100 normal human samples. Overall, the ten cytokine assays were highly sensitive and precise — ideal for clinical research purposes. The future direction is to transfer this application onto an automated 96-well platform for increased throughput.

Bio-Plex Suspension Array Technology

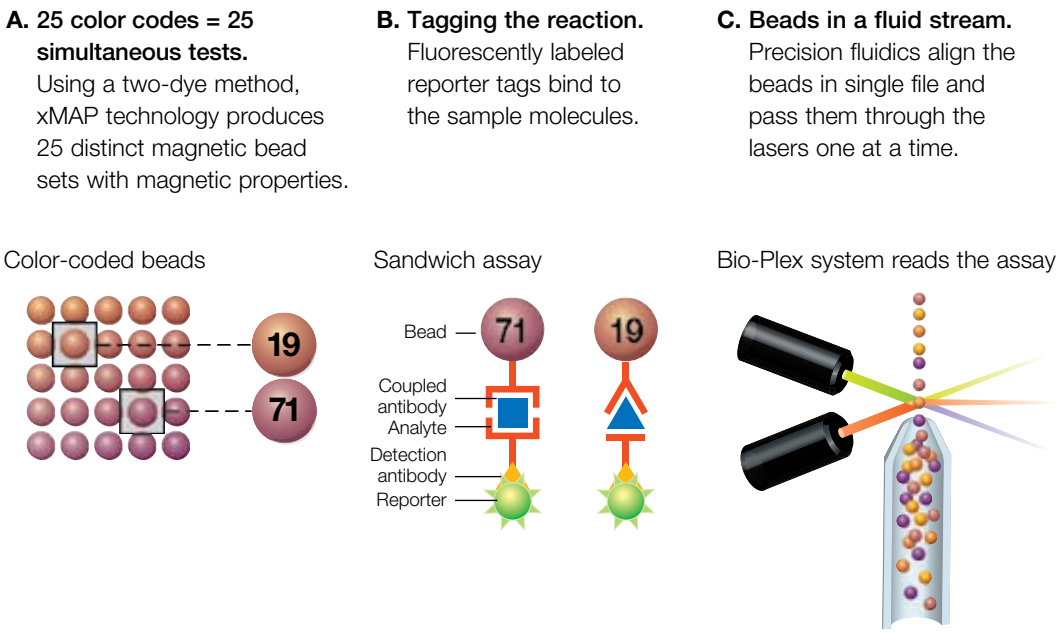


Fig. 1. The Bio-Plex system is a complete suspension array system that comprises a flow-based 96-well fluorescent microplate assay reader integrated with specialized software, automated validation and calibration protocols, and assay kits. Based on Luminex xMAP technology, Bio-Plex magnetic assays can detect up to 25 distinct analytes in as little as 12 μ l of sample.

10-Plex Human Cytokine Standard Curve Profiles

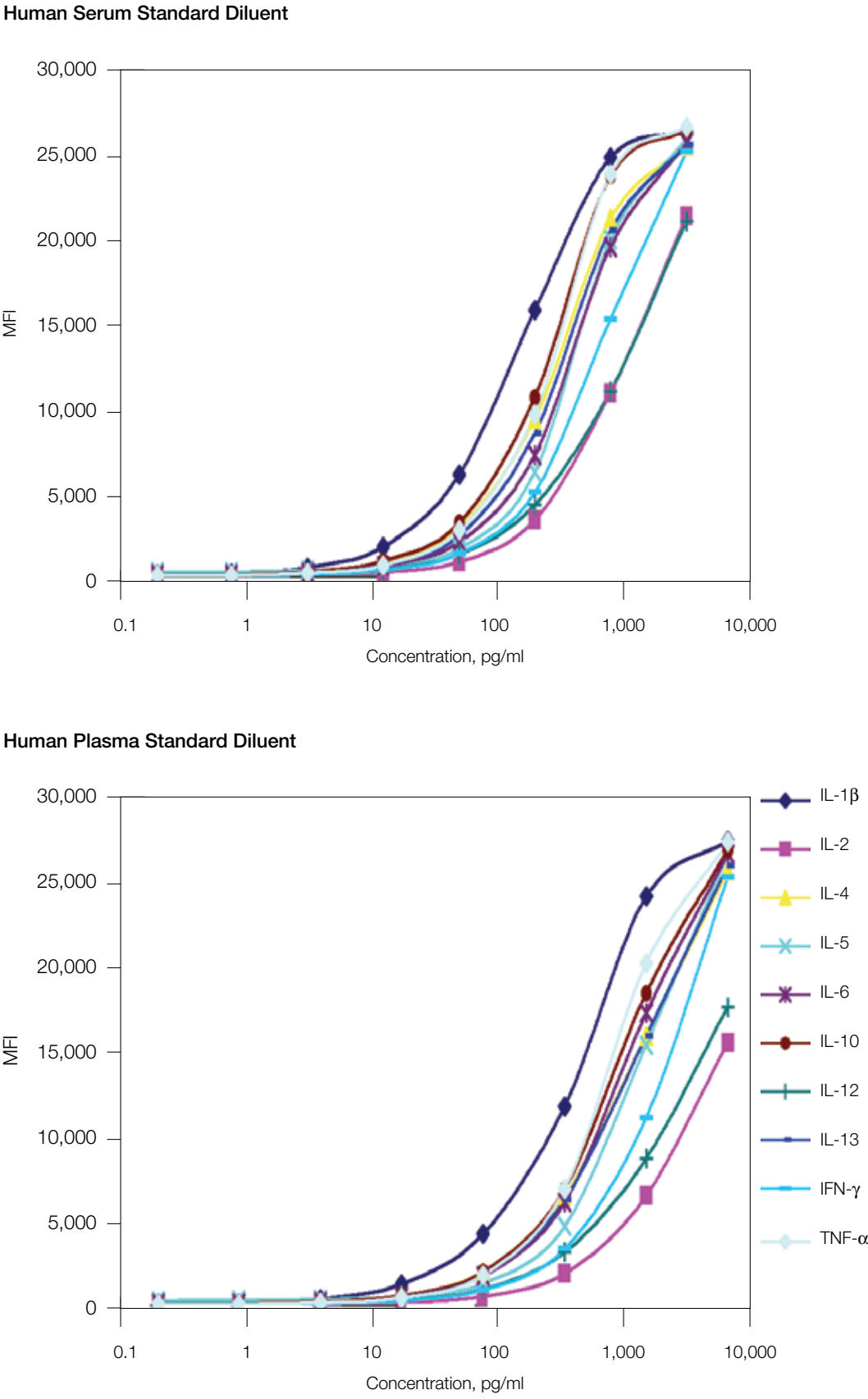


Fig. 2. Five-parameter logistic (5-PL) curve fitting with Bio-Plex Manager™ 4.1 standard software. Shown are median fluorescence intensity (MFI) data for a Bio-Plex Precision Pro™ 10-plex human cytokine assay using human serum and human plasma standard diluents.

Accuracy

Human Serum Pooled Samples Spiked (Technical Specification 80–120% Recovery)										
% Standard Recovery*										
pg/ml	IL-1 β	IL-2	IL-4	IL-5	IL-6	IL-10	IL-12	IL-13	IFN- γ	TNF- α
1,000	114	90	116	90	99	96	93	102	97	95
333	105	94	89	90	90	90	95	90	94	92
111	94	95	96	91	97	98	96	96	97	98
37	99	96	98	88	97	101	94	95	99	97
9.3	98	91	94	85	94	96	92	88	99	96
2.3	94	90	91	73	92	90	97	86	94	90

Human Plasma Pooled Samples Spiked (Technical Specification 80–120% Recovery)										
% Standard Recovery*										
pg/ml	IL-1 β	IL-2	IL-4	IL-5	IL-6	IL-10	IL-12	IL-13	IFN- γ	TNF- α
1,000	125	99	103	101	104	104	94	102	106	111
333	92	97	101	93	97	95	98	95	99	93
111	98	96	104	94	100	100	99	97	100	102
37	103	99	103	95	97	103	99	95	101	103
9.3	106	100	107	96	100	106	98	100	103	107
2.3	98	94	102	67	104	100	88	97	99	105

* Mean of five assays, duplicates of each sample.

Table 1. Human serum or plasma samples from individual donors were pooled (n = 10 each) and spiked with a known amount of 10-plex cytokines. The % spike recovery was calculated from the ratio of measured/expected cytokine concentration in each sample. The mean % spike recovery was determined from five independent assays.

Inter-Assay Precision

Human Serum Pooled Sample Spiked (Technical Specification $\leq 15\%$ CV)										
Inter-Assay %CV*										
pg/ml	IL-1 β	IL-2	IL-4	IL-5	IL-6	IL-10	IL-12	IL-13	IFN- γ	TNF- α
1,000	25.8	9.5	15.4	8.0	1.09	10.7	4.2	10.9	6.4	6.6
333	10.2	7.1	6.7	7.7	7.6	2.4	5.8	9.3	8.0	4.6
111	6.8	1.7	7.8	4.1	4.1	5.3	5.4	2.9	3.7	10.1
37	9.3	6.0	9.9	6.1	6.4	8.4	8.9	7.4	9.7	8.3
9.3	5.2	9.7	2.1	4.9	5.1	4.8	2.0	5.1	8.0	3.0
2.3	6.7	14.6	8.6	6.3	9.1	8.7	6.9	7.3	4.2	10.4

Human Plasma Pooled Sample Spiked (Technical Specification $\leq 15\%$ CV)										
Inter-Assay %CV*										
pg/ml	IL-1 β	IL-2	IL-4	IL-5	IL-6	IL-10	IL-12	IL-13	IFN- γ	TNF- α
1,000	12.3	3.6	7.8	10.7	7.3	7.3	3.9	5.0	6.3	14.8
333	5.3	6.8	8.2	5.5	5.5	6.5	5.2	3.3	3.4	6.4
111	4.9	5.6	10.1	1.6	2.7	2.8	4.4	1.7	2.3	1.9
37	4.9	2.6	5.3	2.8	1.8	4.1	1.7	0.8	3.9	3.0
9.3	6.1	5.7	9.6	10.0	9.6	6.1	5.4	6.4	4.7	9.1
2.3	9.2	17.0	7.2	33.6	10.0	3.7	8.9	10.3	12.9	7.9

* Mean of five assays, duplicates of each sample.

Table 2. Human serum or plasma standard diluents were used to prepare standard curves. Human serum or plasma samples from individual donors were pooled (n = 10 each) and spiked with a known amount of 10-plex cytokines. Inter-assay precision (%CV) was determined from the measured concentrations of 5 independent assays.

Assay Range

	Human Serum Standard (pg/ml)					Human Plasma Standard (pg/ml)				
	LLOQ*	ULOQ**				LLOQ*	ULOQ**			
IL-1 β	0.2	520				0.2	560			
IL-2	1.9	3,200				2.3	3,200			
IL-4	0.4	1,707				0.3	3,200			
IL-5	2.7	3,200				3.7	3,200			
IL-6	0.3	2,760				0.9	3,200			
IL-10	0.2	2,320				0.3	3,200			
IL-12	0.5	3,200				1.4	3,200			
IL-13	0.2	2,320				0.4	3,200			
IFN- γ	0.9	3,200				1.2	3,200			
TNF- α	0.2	1,880				0.2	2,720			

* LLOQ = lower limit of quantitation.

** ULOQ = upper limit of quantitation.

Table 3. Assay range is the concentration range in which the assay is both precise (intra $\leq 10\%$ CV; inter $\leq 15\%$ CV) and accurate (80–120% standard and spike recovery).

Sensitivity

Human Serum Standard (Technical Specifications ≤ 1 pg/ml)							Human Plasma Standard (Technical Specifications ≤ 1 pg/ml)						
Target	Assay/LOD (pg/ml)					Mean LOD	Target	Assay/LOD (pg/ml)					Mean LOD
	1	2	3	4	5			1	2	3	4	5	
IL-1 β	0.11	0.13	0.05	0.07	0.08	0.08	IL-1 β	0.11	0.12	0.10	0.06	0.13	0.10
IL-2	0.22	0.58	0.70	0.39	0.14	0.41	IL-2	0.75	0.78	0.12	0.84	0.69	0.64
IL-4	0.10	0.08	0.16	0.03	0.08	0.09	IL-4	0.20	0.26	0.12	0.10	0.19	0.17
IL-5	1.36	1.41	2.91	0.60	0.54	1.37	IL-5	1.53	0.18	0.20	2.42	0.09	0.88
IL-6	0.38	0.32	0.60	0.25	0.16	0.34	IL-6	0.11	0.42	0.62	0.28	0.04	0.29
IL-10	0.15	0.15	0.15	0.08	0.12	0.13	IL-10	0.08	0.29	0.21	0.14	0.12	0.17
IL-12	0.09	0.11	0.61	0.08	0.32	0.24	IL-12	0.45	0.36	0.52	0.31	0.20	0.37
IL-13	0.15	0.20	0.26	0.16	0.07	0.17	IL-13	0.20	0.18	0.16	0.27	0.14	0.19
IFN- γ	0.10	0.32	0.77	0.01	0.15	0.27	IFN- γ	0.33	0.58	0.48	0.16	0.19	0.35
TNF- α	0.13	0.18	0.21	0.06	0.12	0.14	TNF- α	0.17	0.33	0.17	0.17	0.17	0.20

Table 4. Limit of detection (LOD) is defined as the cytokine concentration obtained at background median fluorescence intensity (MFI) plus 2 standard deviations for the human serum and plasma standard curves.

Specificity

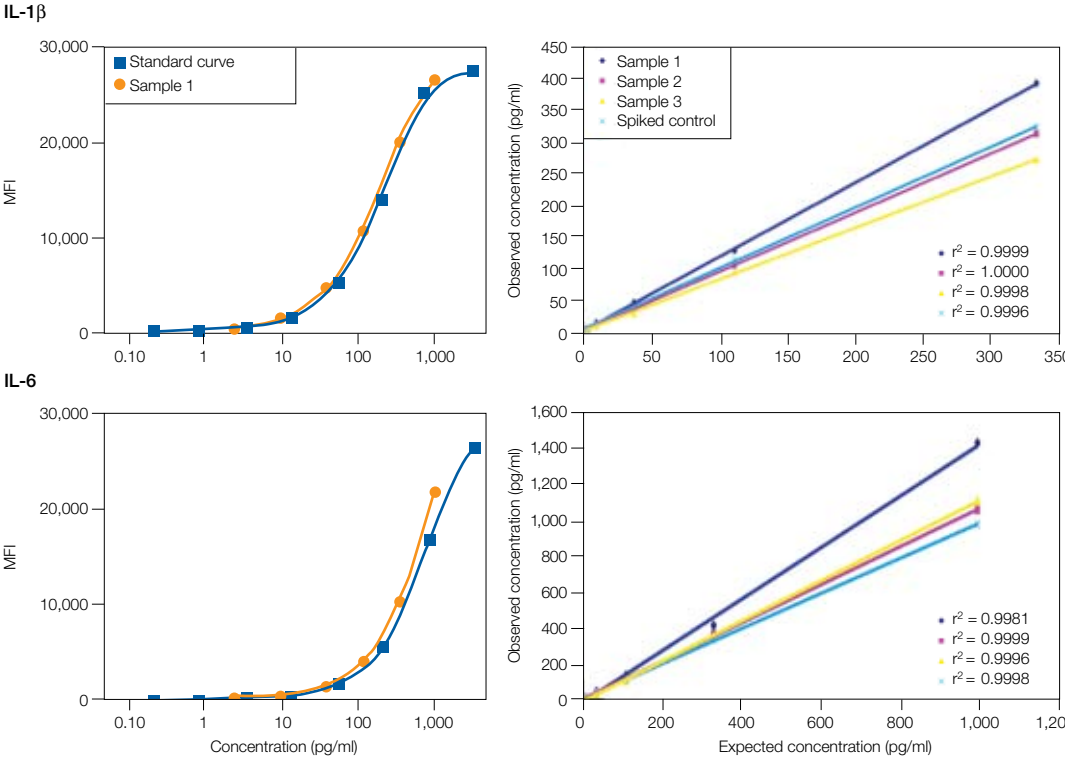
% Cross-Reactivity (Technical Specification $\leq 1\%$ Cross-Reactivity)										
Single Ag		Multiplexed Capture Bead and Detection Ab								
800 pg/ml	IL-1 β	IL-2	IL-4	IL-5	IL-6	IL-10	IL-12	IL-13	IFN- γ	TNF- α
IL-1 β	100.00	0.02	0.00	0.27	0.05	0.00	0.06	0.00	0.03	0.01
IL-2	0.00	100	0.00	0.29	0.05	0.02	0.02	0.00	0.00	0.01
IL-4	0.00	0.09	100.00	0.28	0.13	0.03	0.07	0.01	0.00	0.02
IL-5	0.01	0.03	0.00	100.00	0.04	0.00	0.04	0.00	0.00	0.01
IL-6	0.00	0.04	0.00	0.24	100.00	0.03	0.06	0.01	0.00	0.01
IL-10	0.03	0.04	0.00	0.16	0.12	100.00	0.09	0.00	0.00	0.00
IL-12	0.00	0.02	0.00	0.15	0.06	0.00	100	0.00	0.00	0.00
IL-13	0.00	0.03	0.00	0.25	0.13	0.01	0.07	100.00	0.00	0.01
IFN- γ	0.01	0.07	0.02	0.22	0.08	0.01	0.10	0.01	100.00	0.01
TNF- α	0.00	0.00	0.00	0.14	0.00	0.00	0.00	0.00	0.00	100.00

% Cross-reactivity of Ab_y w /Ag_x = $\frac{\text{FI of Ab}_y \text{ w /Ag}_x - \text{bkgd}}{\text{FI of Ab}_y \text{ w /Ag}_y - \text{bkgd}}$

Table 5. Cross-reactivity was determined from median fluorescence intensity (MFI) of multiplexed capture bead and detection Abs in the presence of single antigens (Ag) at 800 pg/ml.

Parallelism and Linearity

4-PL curves for standard and spiked human serum samples.



4-PL curves for standard and spiked human plasma samples.

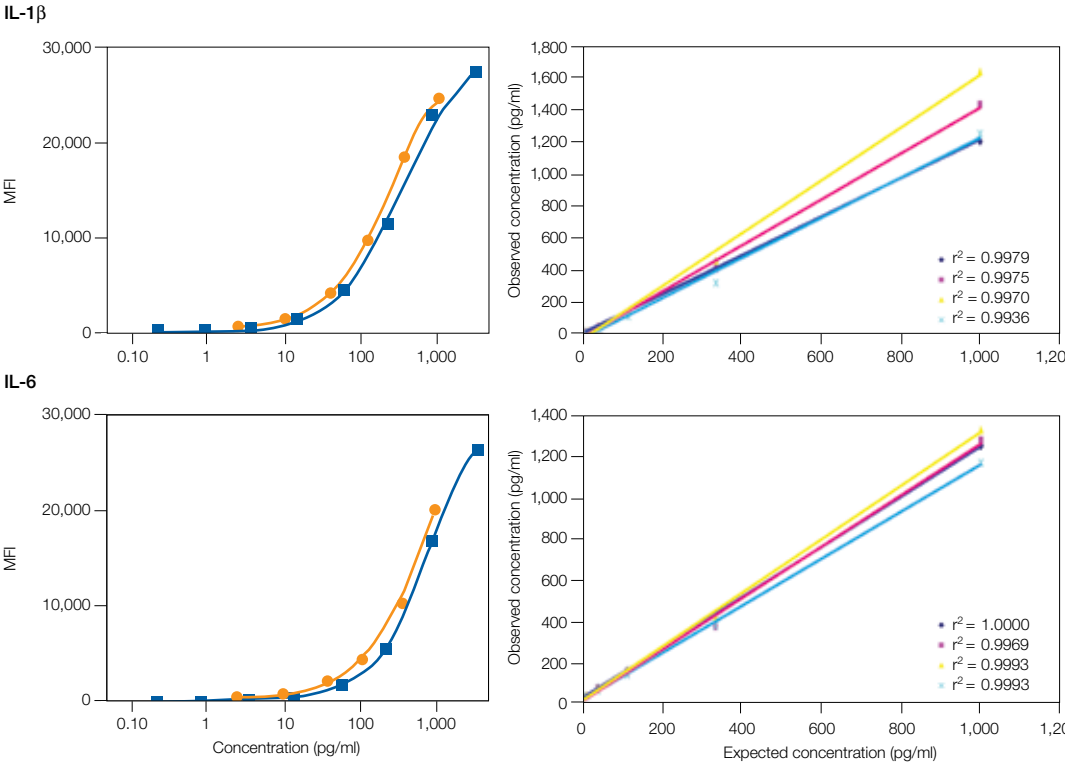


Fig. 3. Examples of parallelism between 8-point standard (blue square) and 6-point spiked serum sample (orange circle) dilutions using 4-PL curve fitting. Slopes were $< 10\%$ different for all cytokines. Linearity was demonstrated in three individual samples with $r^2 > 0.99$ for all ten cytokines.

96-Well Plate Uniformity

	Average % CV of MFI					Ends*	
	96 Wells	Columns 1–12	Rows A–H	Columns	Rows	Columns	Rows
IS	4	3	4	3	4		
IL-1 β	3	3	3	3	4		
IL-2	7	5	7	5	10		
IL-4	6	5	5	6	5		
IL-5	7	4	7	4	11		
IL-6	4	4	3	5	3		
IL-10	3	3	3	3	2		
IL-12	4	3	3	3	3		
IL-13	5	5	4	4	5		
IFN- γ	4	4	4	5	5		
TNF- α	4	4	4	5	5		

* End wells defined as first and last two wells.

Table 6. Summary of 96-well plate uniformity. MFI precision was determined across rows, across columns, and between the first and last two wells of each row and column. IS (internal standard) beads contain a covalently linked fluorophore to measure instrument-related variance.

Cytokine Levels in Normal Human Serum Samples

100 Human Serum Samples (pg/ml)				
	Range			
	Mean	Median	Min	Max
IL-1β	0.30	0.03	0.00	12
IL-2	1.27	0.00	0.00	30
IL-4	0.67	0.00	0.00	18
IL-5	11.09	0.00	0.00	919
IL-6	5.24	0.50	0.00	179
IL-10	0.65	0.14	0.00	15
IL-12	12.52	0.27	0.00	1,053
IL-13	0.58	0.00	0.00	30
IFN-γ	3.66	0.25	0.00	76
TNF-α	1.38	0.16	0.00	41